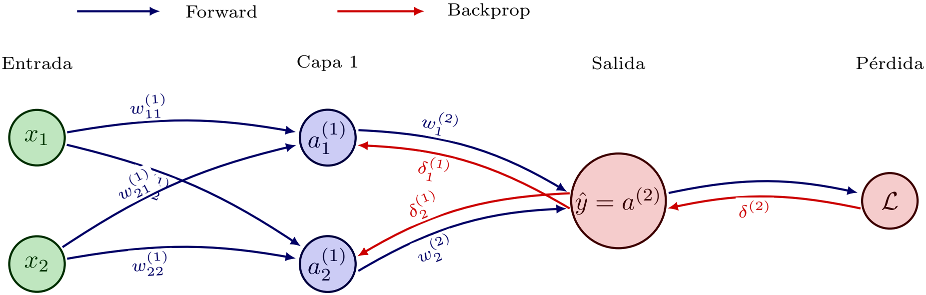

Forward:

$$z^{(1)} = W^{(1)}x + b^{(1)}, \quad a^{(1)} = \sigma(z^{(1)})$$

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}, \quad \hat{y} = a^{(2)}$$

$$W^{(l)} \leftarrow W^{(l)} - \eta \, \nabla_{W^{(l)}} \mathcal{L}$$

Backprop:

$$\delta^{(2)} = \nabla_{a^{(2)}} \mathcal{L} \odot \sigma'(z^{(2)})$$

$$\delta^{(1)} = (W^{(2)})^T \delta^{(2)} \odot \sigma'(z^{(1)})$$

$$\frac{\partial \mathcal{L}}{\partial W^{(2)}} = \delta^{(2)} (a^{(1)})^T, \quad \frac{\partial \mathcal{L}}{\partial W^{(1)}} = \delta^{(1)} x^T$$